NeurIPS Open Polymer Tc Prediction

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1.0 Introduction

Predicting the thermophysical properties of polymers remains a central challenge in chemical engineering and materials informatics, and this study focuses on predicting polymers with high thermal conductivity (Tc) for thermal paste applications. The 2025 NeurIPS Open Polymer Prediction Challenge provides a curated dataset of 7,973 simulated polymers represented in SMILES format and paired with five molecular dynamics (MD)-derived properties, glass transition temperature (Tg), fractional free volume (FFV), thermal conductivity (Tc), density, and radius of gyration (Rg). These properties collectively capture thermal behavior, packing efficiency, segmental mobility, and structural flexibility, making the dataset well-suited for structure–property relationship modeling.

The workflow studies the exploratory data analysis (EDA), feature relevance to Tc prediction, and the performance evaluation of two modelling approaches: Kernel Ridge Regression and XGBoost. Both tuned and untuned model variants were tested to understand how hyperparameter choices influence prediction quality and to determine the performance-maximizing configuration. The study uses SMILES strings and RDKit-generated molecular descriptors, which provide a flexible and interpretable way to represent polymer structures. The overall aim is to build a clear and robust ML pipeline that supports rapid polymer screening and deepens our understanding of the structural factors that influence thermal conductivity in polymer systems.

2.0 Methodology

2.1 Dataset and Summarized Exploratory Data Analysis (EDA)

The EDA revealed four distinct distribution patterns. Tc showed a clear bimodal trend (peaks at ~0.20 and ~0.35 W/m·K), indicating two polymer classes, likely amorphous and semi-crystalline systems. Several RDKit descriptors displayed discrete, step-like distributions due to their categorical nature (e.g., aromatic ring count, H-bond donors), while others such as FFV exhibited very low variance across the dataset. Correlation analysis showed that Tc is strongly linked to rotatable bonds, logP, Rg, and ρ. These nonlinear, multivariate relationships justify the use of kernel-based and tree-boosted models instead of purely linear approaches.

2.2 Data preprocessing and feature engineering

The data preprocessing is demonstrated in the first row (highlighted by green) in Figure 1. Duplicates and empty rows are dropped for different combinations of Tg, FFV, Tc, Rg with RDkit features followed by splitting. It was noticed that each feature from the original data set has a significant number of empty data points. For example, although the full dataset contains 737 data points for the Tc, many of these rows lack values for one or more other features. Directly removing all rows with missing entries would therefore eliminate most of the dataset. To address this, the feature selection was first carried out by filtering the best combination of original data set features with RDkit features. This is done by manually testing all combinations to examine their effect on different preliminary models summarized in Appendix B1. The R^2 reported are from the corresponding test set where the optimum combination is identified as ρ and R_g with RDkit features. This aligns with our EDA observation that ρ and R_g are highly correlated to target. The histogram of Tc from each combination is generated and bimodal distributions are observed from all of them. Therefore, random split and stratified split are tested on the optimum combination in preliminary models with stratified split which obtained a better performance. The threshold of stratified split is chosen as 0.3 based on literature reported optimum range of Tc for thermal paste polymer matrix.[1]

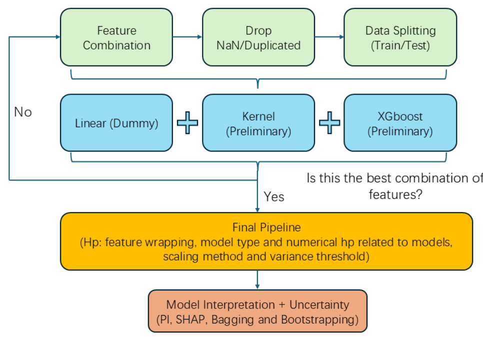

Figure 1: The work flow diagram of model developing.

2.3 Model selection and Training

A linear model is developed as the base line for the model selection. Radical based Kernel's model is selected for its ability to capture non-linear relationships with relatively low computation cost. However, considering its inherent limitation on high dimensional data and long computation time, XGBoost is selected as another candidate for its assembled boosting and bagging features. It also has strong performance on tabular data. Two preliminary models of XGboost and Kernel model are trained with separate pipelines for hyperparameter selection (blue section). This helps with choosing the hyperparameters and their corresponding range for each model in the final pipeline. Additionally, these preliminary models serve as an indicator of what might happen inside the final pipeline and a reference of how well the final pipeline performs.

2.4 Hyperparameter Tuning and Validation Strategy

The hyperparameters selected for KRR and XGBoost in the preliminary model and final pipeline are summarized in appendix B1. The hyperparameters related to KRR and XGBoost and their range are first chosen based on literature[2–4] and tested by manipulating the hyperparameter's range in preliminary models. Once the performance reaches a plateau with a desired performance in the test set (R^2> 0.8). The numerical hyperparameter will be accepted for the final pipeline, where the model is trained and optimized for non-numerical hyperparameters of model type, feature wrapping (within the optimum combination), scaling method and variance threshold. Cross validation is applied as there are only 531 data points after preprocessing and further splitting of a separate validation set would reduce the data available for model training. Random search is applied to reduce the computation time. The number of iterations are also optimized in the final model by increasing the range from 10 to 500 until a plateau shown in appendix A1. Both 5 and 10 folds are tested on the final (n_iteration=500) model and negligible improvement was observed from higher folding. Therefore, 5 folding is applied for lower computation demand and consistency. In the wrapper-based feature selection in the pipeline, several feature subsets consisting of the top target-correlated variables were generated and evaluated. Subsets containing the top 2 through 14 correlated features were assessed, correlation features of both spearman based and pearson based methods are tested without notable difference as the interaction might be sufficient to cover the difference in two methods.

3.0 Result and discussion

3.1 Model Performance interpretation and application result

From the final pipeline, the combination of ρ and Rg, XGBoost model is selected as the most optimized mode with R^2 of 0.959 in the train set, indicating that the model has a high generalisation of the train data set, however the performance of the model has a drop of 0.140 in R^2 on the test set indicating a potential overfit. This overfit is more obvious in the XGBoost model without optimization shown in Table 2, where the train model obtained a R^2 of 1 and a test set of 0.748. This is because a relatively small data set is fitted into XGBoost where the model is capturing not only the true structure–property relationships but also noise and fine-grained patterns specific to the training set. A similar pattern has been reported in literature.[5] Through the hyperparameter optimization, the overfit is significantly reduced with the model maintaining a strong performance on the test set (R^2 = 0.819), demonstrating that it still generalizes reasonably well to unseen data.

Table 1: Performance of different models on their R^2, MAE, Max, MAPE and MSE.

Data Set	**XGBoost R^2 (Final)**	**XGBoost R^2 (untuned)**	**KRR R^2 (tuned)**	**Linear R^2 (Baseline)**
Train	0.959	1	0.875	0.767
Test	0.819	0.748	0.819	0.607
Model	**MAE**	**Max**	**MAPE**	**MSE**
XGboost	0.0216	0.0264	8.98%	4.84E-04
KRR	0.0203	0.0283	8.29%	4.60E-04

The model from the final pipeline is compared to the preliminary models and base line summarised in Table 2. The large difference between linear base-line and other models indicates that there is a nonlinear correlation between the features and the target variable. More precisely, in Table 1 when both ρ and R_g are dropped, the performance of both XGBoost and KRR are similar to linear model with relatively low generalization (R^2 ~0.75), in contrast, when Rg and ρ are included, KRR and XGBoost both out performed linear baseline. This indicates that the features nonlinearly correlated to target are dropped in the first case, which aligns with the EDA report where T_c has high interactive correlation with R_g. It is noticeable that the KRR model has a very close performance of R^2 with XGBoost, moreover, the drop in performance between the train and test set are significantly smaller indicating a more consistent model performance.

This aligns with the optimum performance of the Kernel model in a small data set and low dimension data in literature.[5–7] To further compare the KRR and XGBoost, both models were tested from the Kaggle competition test set summarised in Table 2. The Kaggle test set has only three samples, so R^2 becomes unstable from individual data points that heavily skew it, which may yield negative values, and can indicate poor reliability with extremely small datasets. In such cases, MAE, MSE, MAPE and Max error functions were applied. KRR has slightly better performance in most of the test methods and proved its ability in our data set. It was noticed that XGBoost model outperformed KRR in terms on max error, likely because outliers are not removed during the training, which has more effect on KRR model as it generalizes the data based on the distance between points and outliers will significantly affect its performance in terms of distance to regular data. Conversely, XGBoost as a tree based model can assign a separate leaf for the outlier to minimise the influence on the rest of the model.

The performance of the final optimized model directly aligns with the main research objective to determine whether machine-learning methods can meaningfully predict the T_c of a thermal paste. The test-set R^2 value of 0.863 indicates that the model is able to estimate T_c for new polymers with a high level of reliability using the selected features. Since the model remains consistent across both mid-range and higher T_c values, it can be used to shortlist promising polymers and flag those that may exhibit superior T_c. This makes the workflow useful as an initial screening tool for identifying potential candidates for thermal interface materials or thermal pastes, where efficient heat transfer is essential.

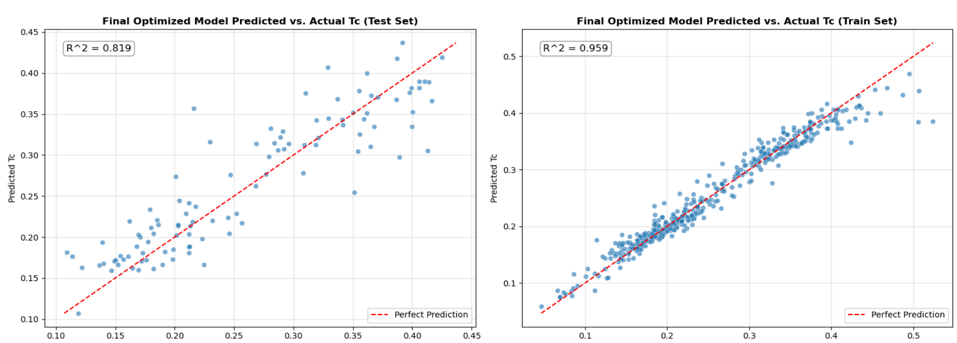

Figure 2: Performance of final pipeline model on train and test set.

3.2 Feature Importance & Uncertainty Analysis

To quantify epistemic uncertainty associated with model prediction, ensemble models were leveraged via bagging (to train random forests on resampled & replaced data) in order to visualize the ranges within the test set that may be under-sampled from a given random seed. Because the XGBoost model generally yielded the most accurate predictions, it served as the basis for the ensemble models. A 95% confidence interval was used to quantify areas of greater ensemble disagreement (Appendix A, Figure A4). The number of bootstraps was arbitrarily chosen to be a high number (n = 1000) based on availability of computational power, and the principle that increased ensembles generally decreases MSE via a linear dependency,[8] despite diminishing returns. Prediction reliability by comparing predicted uncertainties to their actual errors from test data predictions (Appendix A, Figure A5) was denoted by an R^2 of 0.492. The plot was observed to have very few predicted high-uncertainty data points which deflated the correlation value. Feature importance was visualized through the mean difference in model scoring between baseline and post-permutation via R^2, with 10 replicate permutations, reaffirming the high correlations of RotBonds, ρ and R_g (Appendix A, Figure A6).

3.3 Reference and justification to literature

Lin et al. show that RDKit descriptors reliably capture nonlinear structure–property trends and that XGBoost tends to overfit small datasets, directly supporting our feature choice and tuning strategy.[9] Xu et al. confirm that Rg and hydrogen-bonding correlate positively with thermal conductivity and that most amorphous polymers lie in the 0.1–0.5 W/m·K range,[10] aligning with our EDA and dataset distribution. It further shows that ML accelerates high-T_c polymer discovery, validating the usefulness of our final pipeline.

4.0 Conclusion

This study shows that supervised machine-learning models can reliably predict polymer T_c and support early-stage screening of candidates for thermal paste applications. By combining EDA, RDKit descriptors and a systematically tuned pipeline, both XGBoost and KRR achieved strong performance, with test-set R^2 values above 0.8 and low prediction errors. These results confirm that nonlinear structure–property relationships can be captured effectively using the selected features and modelling approach. Future work should focus on expanding the feature space, improving uncertainty quantification, and validating predictions with larger or experimentally derived datasets. Overall, the workflow provides a clear and reproducible foundation for ML-guided polymer design.

Video Link

https://www.youtube.com/watch?v=ORQR66bwtzc

Github Repository Link

https://github.com/kevzhu14/NeurIPS-Open-Polymer-Prediction

Acknowledgements

Google Gemini 2.5 Flash was used in part to generate certain portions of the codebase, such as for all plotting-based functions.

References

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Appendix

Appendix A - Figures

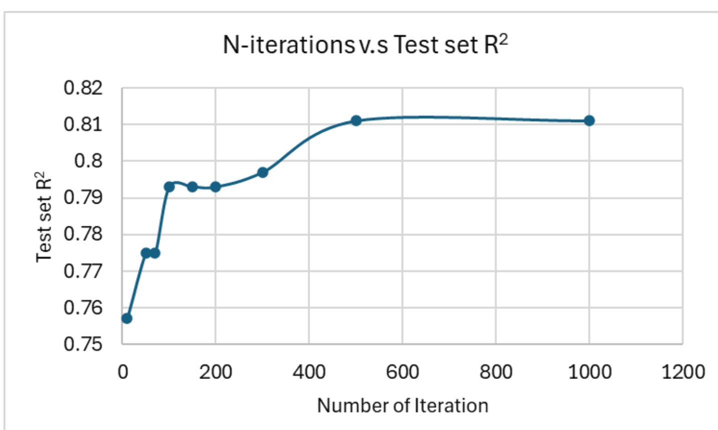

Figure A1: R^2 vs number of iterations for XGBoost model.

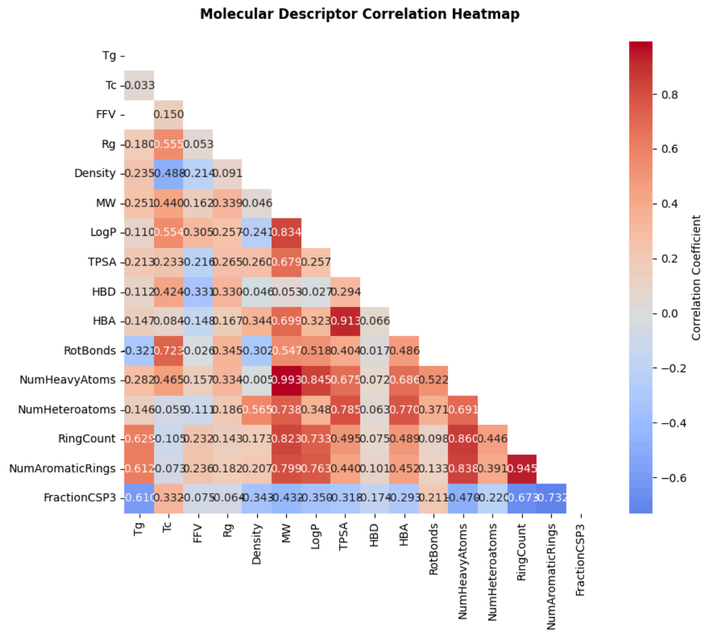

Figure A2: Molecular Correlation Heatmap including all variables

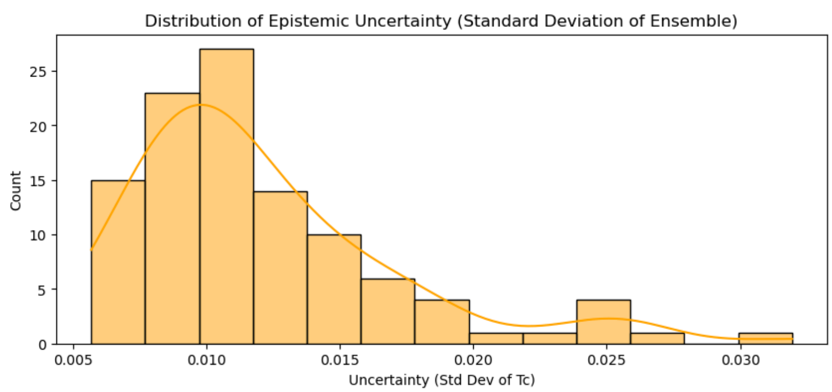

Figure A3: Epistemic Uncertainty Histogram.

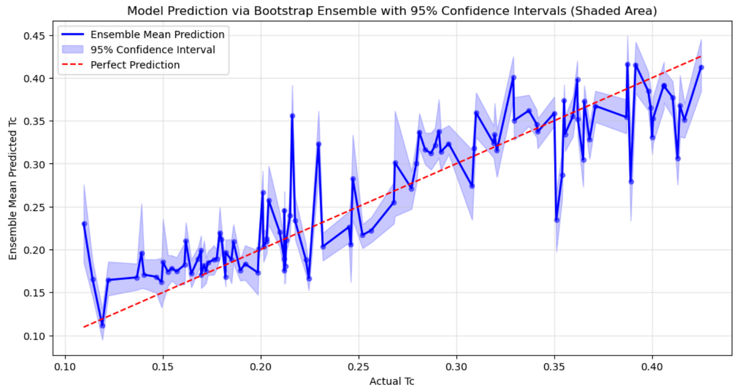

Figure A4: Bootstrap ensembles-informed epistemic uncertainty of prediction using 1000 bootstraps. Shaded areas denote the 95% confidence interval

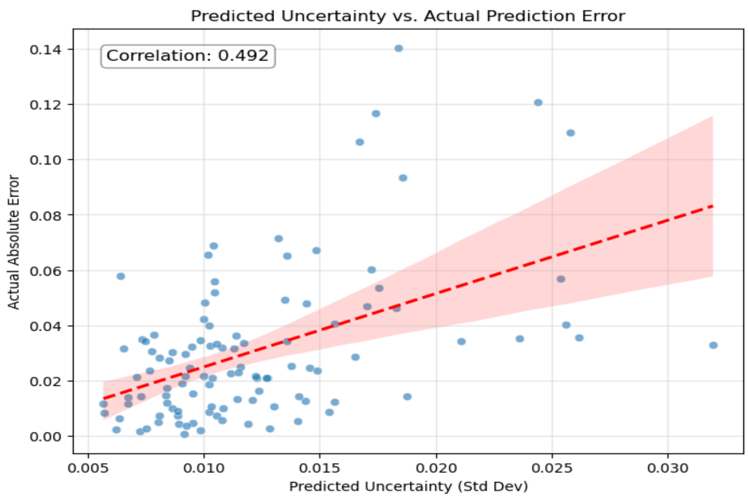

Figure A5: Prediction reliability plot portraying uncertainty prediction vs actual prediction error.

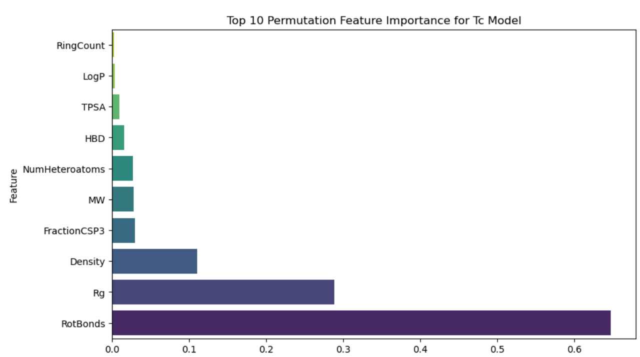

Figure A6: Bootstrap ensembles-informed epistemic uncertainty of prediction using 1000 bootstraps. Shaded areas denote the 95% confidence interval.

Appendix B - Tables

Table B1: R^2 results from different combination of the models

Feature / Splitting Method	XGBoost R^2	KRR R^2	Linear R^2	Number of Rows Post Preprocessing
Only RDkit	0.775	0.75	0.725	737
Rg+RDkit	0.768	0.715	0.598	535
Rg+Density+RDkit	0.778	0.756	0.685	531
FFV+Rg+Density+RDkit	0.756	0.58	0.705	221

Table B2: Absolute Correlation of Tg and Tc with features and targets

S.No	Column	Absolute Correlation	Column	Absolute Correlation
0	Tc	1	Tg	1
1	RotBonds	0.722609	RingCount	0.628617
2	Rg	0.554758	NumAromaticRings	0.612252
3	LogP	0.554276	FractionCSP3	0.610485
4	Density	0.488476	RotBonds	0.320607
5	NumHeavyAtoms	0.464714	NumHeavyAtoms	0.281691
6	MW	0.439685	MW	0.250809
7	HBD	0.424403	Density	0.235291
8	FractionCSP3	0.331863	TPSA	0.212706
9	TPSA	0.233444	Rg	0.180072
10	FFV	0.149878	HBA	0.14733
11	RingCount	0.104799	NumHeteroatoms	0.146351

12	HBA	0.083799	HBD	0.111675
13	NumAromaticRings	0.073421	LogP	0.110265
14	NumHeteroatoms	0.058626	FFV	NaN

Table B3. Hyperparameter selection in the preliminary models

Hyperparameter	Range
variance_treshold__threshold	0.0, 1e-3, 1e-2, 1e-1, 0.5, 1.0
scaling	None,MinMaxScaler(), StandardScaler()
selectkbest__k	2, 3, 4, 5,6,7,8,9,10,11,12,13,14
krr__alpha	np.logspace(-10,10,10)
krr__gamma	np.logspace(-10,10,10)

Hyperparameter	Range
variance_treshold__threshold	0.0, 1e-3, 1e-2, 1e-1, 0.5, 1.0
scaling	None,MinMaxScaler(), StandardScaler()
selectkbest__k	2, 3, 4, 5,6,7,8,9,10,11,12,13,14
xgbregressor__n_estimators	100, 200, 400, 600, 800, 1000
xgbregressor__learning_rate	0.005, 0.01, 0.05, 0.1, 0.2, 0.3

xgbregressor__max_depth	3, 5, 7, 10, 12, 15
xgbregressor__min_child_weight	1, 3, 5, 7, 10, 15
xgbregressor__gamma	0, 0.05, 0.1, 0.2, 0.4, 0.6, 0.8
xgbregressor__subsample	0.5, 0.6, 0.7, 0.8, 0.9, 1.0
xgbregressor__colsample_bytree	0.5, 0.6, 0.7, 0.8, 0.9, 1.0

Table B4: Absolute Correlation of Tg and Tc with selected features and targets

	Tc	Absolute Correlation	Tg	Absolute Correlation
1	RotBonds	0.722609	RingCount	0.628617
2	Rg	0.554758	NumAromaticRings	0.612252
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7	RingCount	0.104799	Rg	0.180072
8	NumAromaticRings	0.073421	LogP	0.110265